

From a Suspicious Silence on the ECG to an Interpretable Classifier: Detecting Brugada Syndrome on the Brugada-HUCA Dataset

An independent research project — problem, experimentation, model, and findings

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Abstract

Brugada syndrome (BrS) is an inherited heart-rhythm disorder that can cause sudden cardiac death in young, otherwise healthy people, often without any warning symptoms [1]. Its diagnostic electrocardiogram (ECG) signature — ST-segment elevation in the right precordial leads — is frequently subtle, intermittent, or entirely absent on a resting ECG, and reliably unmasking it currently requires a provocative drug challenge that itself carries a small but real proarrhythmic risk [3,4]. This project began as a personal effort to understand why that diagnostic gap exists and whether a transparent, non-intrusive computational tool could help narrow it. Using the newly released Brugada-HUCA dataset (PhysioNet v1.0.0, February 2026; 76 BrS patients, 287 controls), I built an interpretable machine-learning classifier from clinically grounded ECG features — J-point and ST-segment amplitudes, QRS duration, and T-wave morphology extracted from robust per-lead median beats — and validated it with patient-level repeated stratified cross-validation. The resulting XGBoost model achieves an AUROC of 0.90 (95% CI 0.80–0.97) and 76% sensitivity at a fixed 90% specificity. SHAP attribution confirms the model relies on the expected right-precordial (V1–V2) ST/J-point signal rather than an incidental artifact. Critically, the error analysis shows the model’s failures are not random: sensitivity is 62% when the baseline ECG is non-pathological versus 83% when it is overtly abnormal, and missed cases sit at an intermediate degree of ST elevation between controls and correctly identified patients. This mirrors, in miniature, the exact clinical problem that motivated the project. The work is presented honestly as a single-center, small-sample research baseline — not a validated diagnostic tool — and is benchmarked against prior BrS-AI literature to place it in context rather than to claim priority over it.

1. Introduction

1.1 Why I started this

I wanted this to be more than an exercise in fitting a model to a dataset. Brugada syndrome kept coming up in my reading as a strange kind of medical problem: a person can be young, active, and feel completely fine, and the only sign that something is wrong is a specific shape in a heartbeat that a resting ECG might not even show that day. The condition is treatable once found — but finding it is the hard part. That gap between “the disease is real and dangerous” and “the test for it doesn’t reliably show it” is what pulled me in, and it is the problem this whole project is organized around: first understanding *why* the gap exists, then experimenting with whether a transparent model could help narrow it, and finally building and honestly evaluating that model.

1.2 The clinical problem

Brugada syndrome is an inherited cardiac channelopathy with an estimated prevalence around 1 in 2,000, occurring more often in young to middle-aged men, and it predisposes patients to ventricular fibrillation and sudden cardiac death in an otherwise structurally normal heart [1]. Roughly one in ten cases of sudden cardiac death is caused by an inherited arrhythmic condition such as BrS [6]. About 20% of clinically diagnosed probands carry an identifiable mutation in *SCN5A*, the gene encoding the main cardiac sodium channel, though the majority of cases have no single identified causative variant, which is itself a reminder that BrS is defined electrocardiographically, not purely genetically [1,2].

The diagnostic hallmark is ST-segment elevation in the right precordial leads (V1–V2): a **type-1 (“coved”)** pattern with at least 2 mm of J-point elevation and a downsloping ST segment is diagnostic on its own, while a **type-2 (“saddleback”)** pattern is suggestive but requires further evaluation [1]. The problem is that this pattern is dynamic. It can be present on one ECG and absent on the next, and it is known to be suppressed or unmasked by fever, autonomic tone, certain medications, and normal day-to-day physiological variation [1,5]. This is precisely what makes the resting ECG an unreliable gatekeeper: a single normal-looking tracing does not rule out the disease.

1.3 Why the current fix is imperfect

When a resting ECG is inconclusive, the established next step is a sodium-channel-blocker provocation test (commonly ajmaline or flecainide), which pharmacologically unmasks the type-1 pattern in patients whose baseline ECG is otherwise silent. In one large cohort of families evaluated after a sudden unexplained death, ajmaline testing was required to reach the diagnosis in 97% of the individuals ultimately found to have BrS — the resting ECG alone would have missed nearly all of them [3]. That same study found that 17% of patients with an initially “concealed” pattern went on to develop a spontaneous type-1 pattern or a clinically significant arrhythmic event on follow-up, underscoring that concealed BrS is not a lesser diagnosis, it is a harder-to-see one [3]. The catch is that provocation testing, while generally safe in expert hands, is an invasive pharmacological stress test with a recognized (if small) risk of inducing the very ventricular arrhythmias it is trying to unmask, so it is reserved for patients where clinical suspicion already justifies the risk [4,5]. There is no established, non-intrusive way to flag which resting ECGs deserve that extra scrutiny in the first place. That is the specific opening for a computational tool: not to replace provocation testing or genetic counseling, but to help triage which unremarkable-looking ECGs might be worth a second look.

1.4 What others have already tried

This project is not the first attempt to apply machine learning to this problem, and it should not be read as though it were. A deep-learning model trained on ECG waveforms without provocation testing achieved 88.4% accuracy and 0.934 AUC in validation on an Italian cohort, using a convolutional approach on raw, aligned beats [6]. A separate study restricted itself entirely to P-wave morphology — reflecting an atrial phenotype associated with BrS rather than the classic ventricular ST/J-point signature — and reached 81% accuracy (86% sensitivity, 73% specificity) with an AdaBoost classifier on 123 patients [7]. More recently, a self-supervised pretraining approach (VICReg) was used specifically to address the label scarcity that plagues rare-disease ECG classification, reaching an AUROC of 0.88 and AUPRC of 0.82, and was further used to flag previously missed BrS cases in an institutional cohort, suggesting real-world BrS prevalence may be underestimated [8]. A recent review surveys this growing body of AI-BrS work and notes that AI models have in several settings matched or exceeded trained cardiologists at detecting subtle BrS patterns, while also being used to help classify ambiguous genetic variants [9]. These are legitimate, more mature efforts than this project, and Section 5 returns to them for context rather than competition.

1.5 The gap this project fills, and its actual contribution

What none of the above use is the **Brugada-HUCA** dataset — a 12-lead ECG dataset from Hospital Universitario Central de Asturias, released on PhysioNet in February 2026 [10,11]. At the time of writing, I could not find any published classifier built on it; only the dataset’s own descriptor existed. That gap is what made this a reasonable independent project to take on: not to out-perform the literature above, but to build the *first* documented baseline on this specific, newly available resource, using a transparent, feature-based approach rather than a black-box deep network, so that every part of the model’s reasoning could be checked against known cardiology rather than trusted blindly.

The concrete contributions of this work are: (i) the first independent, fully reproducible classifier on Brugada-HUCA, built from interpretable ECG features with leakage-free patient-level validation; (ii) confirmation, via SHAP attribution, that the model’s decisions are anchored in the clinically expected right-precordial ST/J-point signature rather than incidental artifacts; and (iii) an error analysis that ties the model’s failures directly to the concealed-BrS phenotype described in Section 1.3 — i.e., the model struggles on exactly the patients that provocation testing exists to catch, which is itself a small, self-contained piece of evidence about where the real diagnostic difficulty lives.

2. Data

Table 1. Brugada-HUCA dataset summary.

Property	Value
Source	Brugada-HUCA, PhysioNet v1.0.0 (Feb 2026) [11]
Subjects	363 (76 BrS, 287 healthy controls)
BrS subtypes	69 type-1, 7 type-2
Prevalence	20.9%
Leads	12 standard (I, II, III, aVR, aVL, aVF, V1–V6)
Sampling rate	100 Hz
Duration	12 s (1,200 samples per lead)
Records per patient	1 (confirmed — no train/test leakage risk)
Metadata fields	<code>patient_id</code> , <code>basal_pattern</code> , <code>sudden_death</code> , <code>brugada</code>

The dataset is clean by construction: a single, consistent lead ordering, no missing samples, and exactly one ECG per patient, which means any patient-level cross-validation split is automatically leakage-free. The `brugada` label takes three values (0 = control, 1 = type-1, 2 = type-2); the two BrS subtypes are collapsed into a single binary BrS-positive class for the primary classification task. The `basal_pattern` flag — whether the baseline resting ECG already looks pathological to a reader — is deliberately withheld from the model and used only afterward, in the error analysis, since using it as a feature would trivialize the exact problem the model is meant to address.

Figure 1 shows a representative right-precordial comparison: a confirmed type-1 BrS case (patient 188981) against a control (patient 251972), both drawn directly from the raw dataset.

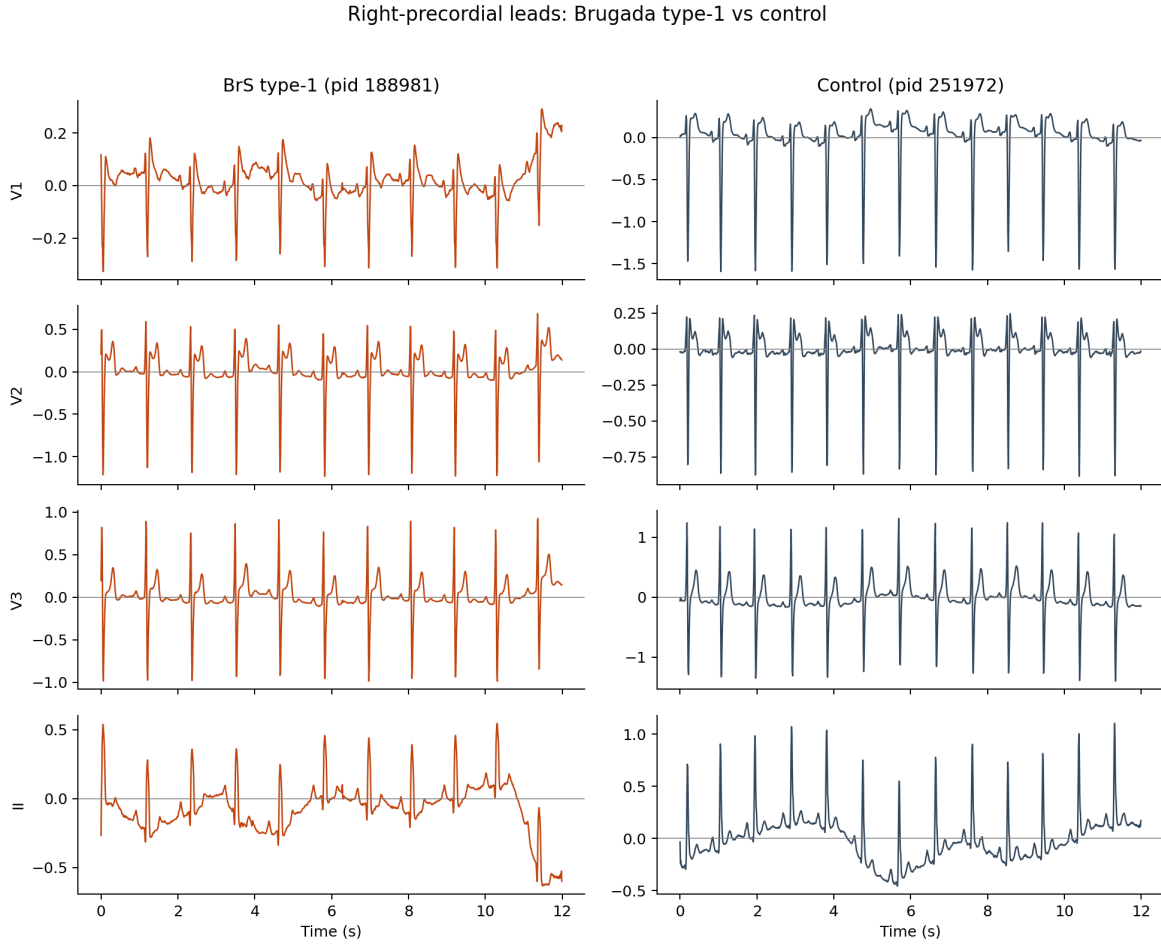


Figure 1: Right-precordial leads for a confirmed Brugada type-1 case versus a control. The BrS case shows the coved, elevated ST/J-point morphology in V1–V2 characteristic of the type-1 pattern, with a downsloping return to baseline; the control shows a sharp, unremarkable return to baseline in the same leads.

3. Methods — the experimentation

3.1 Preprocessing

Every lead was band-pass filtered (0.5–40 Hz, 4th-order Butterworth, zero-phase `filtfilt`) to remove baseline wander and high-frequency noise, and R-peaks were detected on lead II. Quality control was clean across the entire dataset: zero flat leads, zero R-peak-detection failures, and all derived heart rates were physiologically plausible (39–182 bpm, median 69 bpm). No records were dropped. Figure 2 shows this pipeline end-to-end for the same two patients as Figure 1: raw versus filtered signal, the component removed by filtering, and the detected R-peaks.

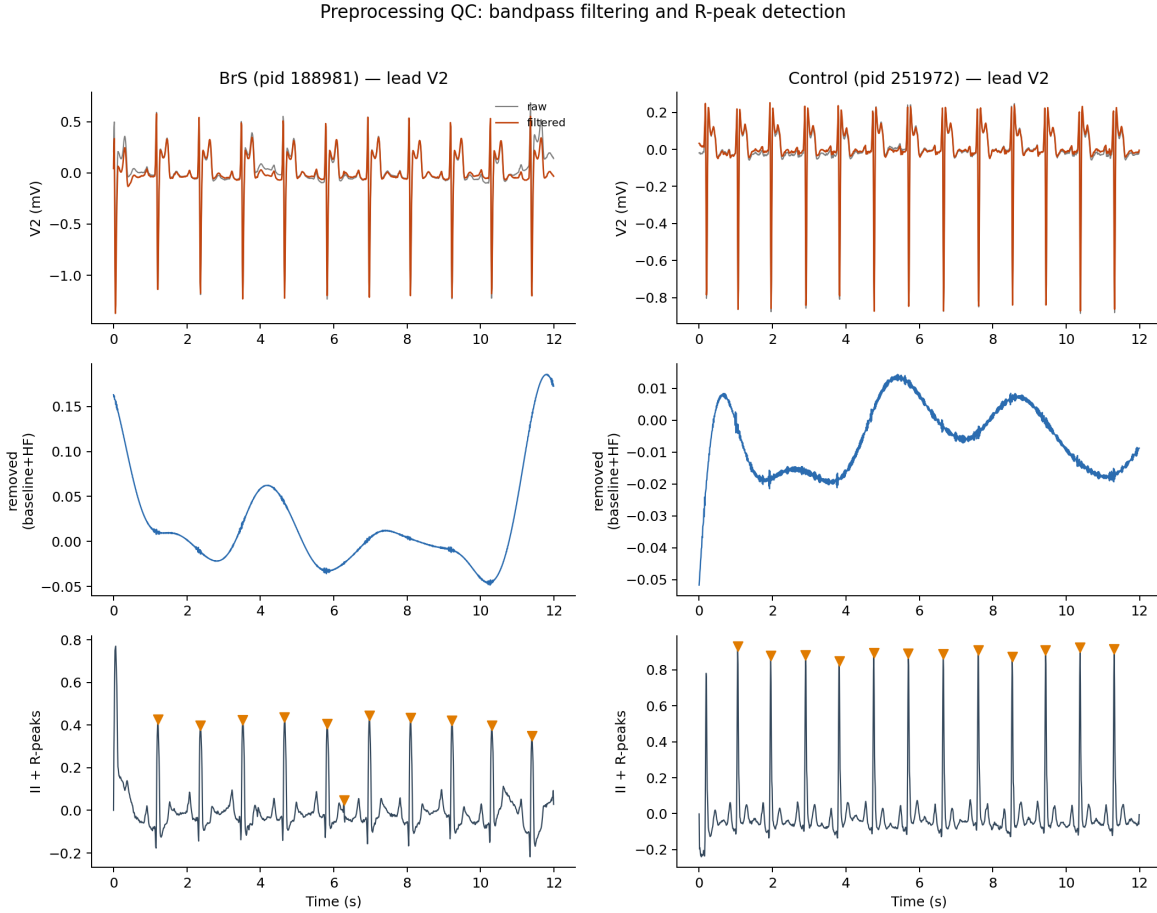


Figure 2: Preprocessing quality control. Top: raw versus band-pass-filtered lead V2. Middle: the removed baseline-wander and high-frequency component. Bottom: detected R-peaks (orange markers) overlaid on filtered lead II. Left column: BrS case; right column: control.

3.2 Feature engineering

For every subject and lead, I built a **robust median beat**: all detected beats for that lead were aligned on the R-peak and combined sample-by-sample using the median, which suppresses single-beat noise and motion artifacts without needing to hand-pick a “clean” beat. From each lead’s median beat I extracted nine interpretable features: J-point amplitude, ST amplitude at J+40 ms and J+80 ms, ST slope, R and S amplitudes, QRS duration, signed T-wave amplitude, and the

J-to-R amplitude ratio. Across 12 leads that gives 108 features, plus three global rhythm features (median heart rate, median RR interval, RR variability), for **111 features total**, with no missing values.

Before any modeling, I screened every feature for univariate discrimination between BrS and control using Cohen’s d and the Mann–Whitney U test with Bonferroni correction. This step matters for two reasons: it is a sanity check on the dataset before trusting any model built from it, and it produces an independent, model-free confirmation of what a downstream classifier *should* find important if it is learning real physiology rather than noise. **43 of 111 features survived Bonferroni correction.** The strongest discriminators, shown in Table 2 and Figure 3, are exactly where BrS pathology is expected to live: the right precordial leads.

Table 2. Top discriminating features (Cohen’s d , Bonferroni-significant).

Feature	Cohen’s d	Interpretation
V2 ST (J+40 ms)	+1.08	right-precordial ST elevation
V1 ST (J+40 ms)	+1.00	right-precordial ST elevation
V4 QRS duration	+0.90	QRS prolongation
V3 QRS duration	+0.90	QRS prolongation
V2 QRS duration	+0.87	QRS prolongation
V2 T-wave amplitude	−0.85	attenuated positive T-wave
V2 J-point amplitude	+0.84	J-point elevation

One nuance worth stating precisely: the V2 T-wave effect is an *attenuation* of a normally positive T-wave in BrS (median +0.18 mV in BrS vs. +0.31 mV in controls), not a group-level inversion to a negative T-wave — only around 28% of BrS records in this dataset show a truly negative V2 T-amplitude. Reporting it as a flat “T-wave inversion” would overstate the effect.

3.3 Model and validation

I trained **XGBoost** as the primary model (300 trees, max depth 3, learning rate 0.05, with `scale_pos_weight` set to correct for class imbalance) and a **Random Forest** as a comparator (`class_weight='balanced'`). Both were validated with **repeated stratified 5-fold cross-validation, 10 repeats (50 folds total)**. Because Brugada-HUCA has exactly one ECG per patient, every one of these splits is automatically patient-level and leakage-free — there is no risk of the same patient appearing in both a training fold and its corresponding test fold. Class imbalance (21% prevalence) was handled through class weighting rather than oversampling, since oversampling methods like SMOTE synthesize new points from nearest neighbors and, done naively across cross-validation folds, can leak information between train and test partitions.

I report AUROC, AUPRC, sensitivity, specificity, and sensitivity at a fixed 90% specificity. Accuracy is reported for completeness but de-emphasized as a headline metric, since a classifier that simply predicts “control” for everyone already scores 79% accuracy on this dataset without being useful at all.

Top discriminating features: BrS vs control (right-precordial signature)

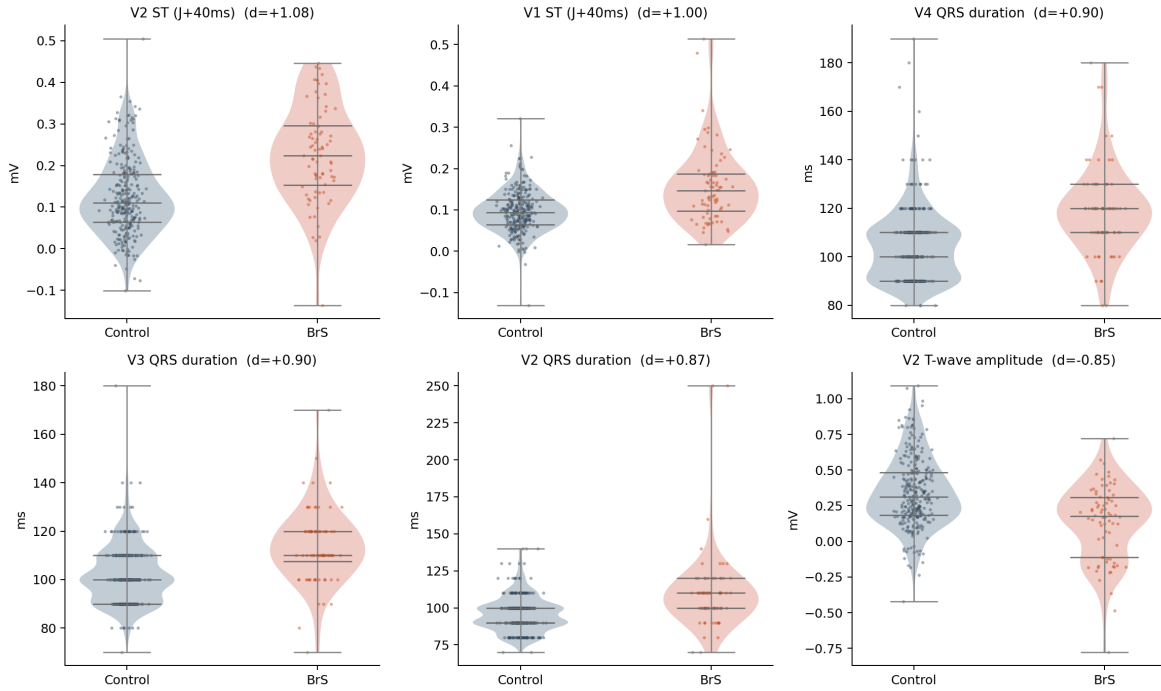


Figure 3: Distributions of the six strongest discriminating features. Each panel shows Control versus BrS with individual patients jittered over a violin density estimate; horizontal lines mark the 25th/50th/75th percentiles. All six line up with the established right-precordial BrS signature: ST elevation and J-point elevation in V1–V2, QRS prolongation across V2–V4, and attenuated T-wave amplitude in V2.

4. Results — the tool

4.1 Classification performance

Table 3. Cross-validated performance (mean over 50 folds, 95% interval in brackets).

Metric	XGBoost (primary)	Random Forest
AUROC	0.900 [0.80, 0.97]	0.910 [0.84, 0.97]
AUPRC	0.777 [0.57, 0.92]	0.789 [0.64, 0.91]
Sensitivity (thr 0.5)	0.693 [0.42, 0.87]	0.725 [0.47, 0.88]
Specificity (thr 0.5)	0.944 [0.88, 1.00]	0.923 [0.88, 0.98]
Sensitivity @ 90% specificity	0.763 [0.53, 0.94]	0.751 [0.53, 0.93]
Accuracy	0.891	0.881

The two models are statistically indistinguishable on AUROC; XGBoost was kept as the primary model for better probability calibration (Figure 4, right panel). As an independent check on these numbers, I re-ran the full cross-validation procedure end-to-end from the raw data checkpoints while preparing this manuscript, which reproduced an AUROC of 0.900 [0.80, 0.97] and AUPRC of 0.775 [0.59, 0.92] — a near-exact match to the original run, and reassurance that the result is not an artifact of one lucky random seed.

The headline clinical trade-off is visible directly in Table 3: at the default 0.5 threshold, the model is conservative (94% specificity) but misses roughly 30% of true BrS cases. Moving the threshold to hold specificity at a fixed 90% recovers sensitivity to 76%. Neither operating point is “the” answer — the right threshold depends on what a missed case versus a false alarm actually costs in a given clinical workflow, which is a decision this project deliberately leaves open rather than presuming.

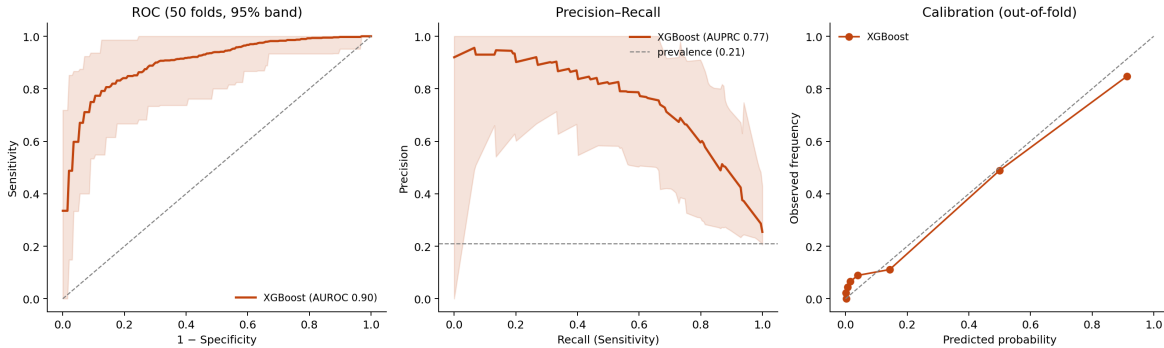


Figure 4: Left: ROC curve with a 95% band across the 50 cross-validation folds (AUROC 0.90). Center: precision–recall curve against the 21% prevalence baseline (AUPRC 0.77). Right: out-of-fold calibration — predicted probabilities track the diagonal reasonably closely, meaning a predicted probability of, say, 0.5 corresponds to an observed BrS rate close to 50% in that bin.

4.2 Interpretability: is the model actually looking at the right thing?

A model can hit a good AUROC for the wrong reasons — by exploiting a scanner artifact, a recording-site quirk, or some other shortcut that happens to correlate with the label in this particular dataset. This is the single most important check for a clinical ECG classifier, so I ran SHAP

(SHapley Additive exPlanations) attribution on the fitted XGBoost model to see which features it actually relies on.

The result lines up with Section 3.2’s univariate screen almost feature-for-feature: the top three features by mean absolute SHAP value are **V2 ST amplitude (J+40 ms)**, **V1 ST amplitude (J+40 ms)**, and **V2 J-point amplitude** — the exact right-precordial signal that defines BrS clinically (Section 1.2). Of the top ten features by SHAP importance, more than half come from leads V1–V3. This consistency between a purely statistical, model-free screen (Table 2) and a trained model’s actual decision attribution (Figure 5) is, to me, the strongest single piece of evidence in this project: it means the model learned the textbook Brugada signature, not an incidental pattern specific to this one hospital’s equipment or patient mix.

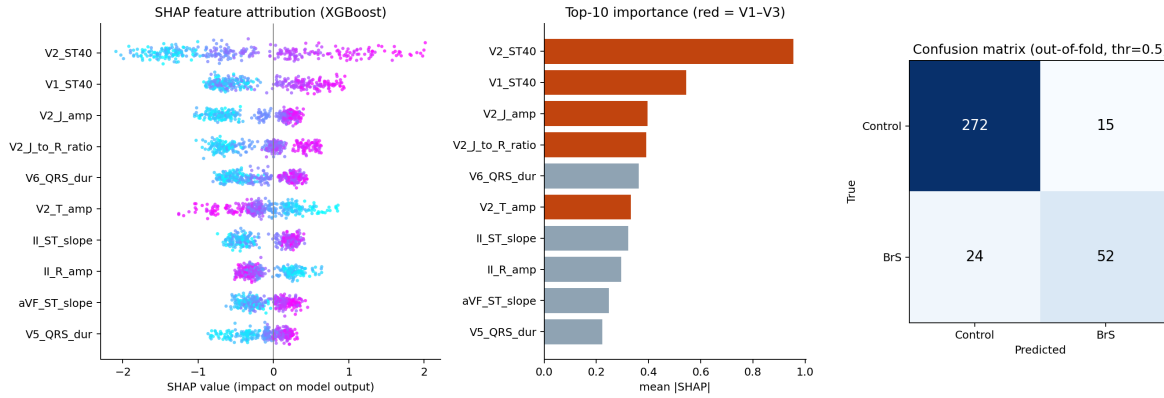


Figure 5: Left: SHAP beeswarm plot — each point is one patient, color encodes that patient’s feature value (magenta = high, cyan = low), and horizontal position shows how much that feature pushed the prediction toward or away from BrS. Center: the same ten features ranked by mean absolute SHAP value; red bars mark right-precordial leads V1–V3. Right: out-of-fold confusion matrix at the default threshold — 272 true negatives, 52 true positives, 24 false negatives, 15 false positives.

4.3 Error analysis: where the model actually fails, and why

Interpretability confirms the model is looking in the right place; the error analysis asks what happens when it still gets it wrong. The 24 false negatives are not spread randomly across the BrS group — they cluster tightly in the **concealed phenotype**. Sensitivity is **62% (33/53)** when the patient’s baseline ECG is non-pathological, versus **83% (19/23)** when the baseline ECG is already overtly abnormal. Subtype sensitivity was 67% (46/69) for type-1 and an anecdotal 6/7 for type-2 — a number too small to draw any real conclusion from, and reported here only for completeness.

Digging further, the missed cases sit at an **intermediate** degree of right-precordial ST elevation: mean V2 ST amplitude of 0.174 mV in missed (false-negative) BrS cases, compared to 0.119 mV in controls and 0.252 mV in correctly identified (true-positive) BrS cases. In other words, the model is not failing randomly — it is failing precisely on the patients whose resting ECG does not fully express the Brugada signature, which is the same population that a routine clinical ECG read would also struggle with, and the same population that sodium-channel-blocker provocation testing exists to catch (Section 1.3). The model’s error pattern is, in a small and self-contained way, evidence for the clinical picture that motivated this project in the first place.

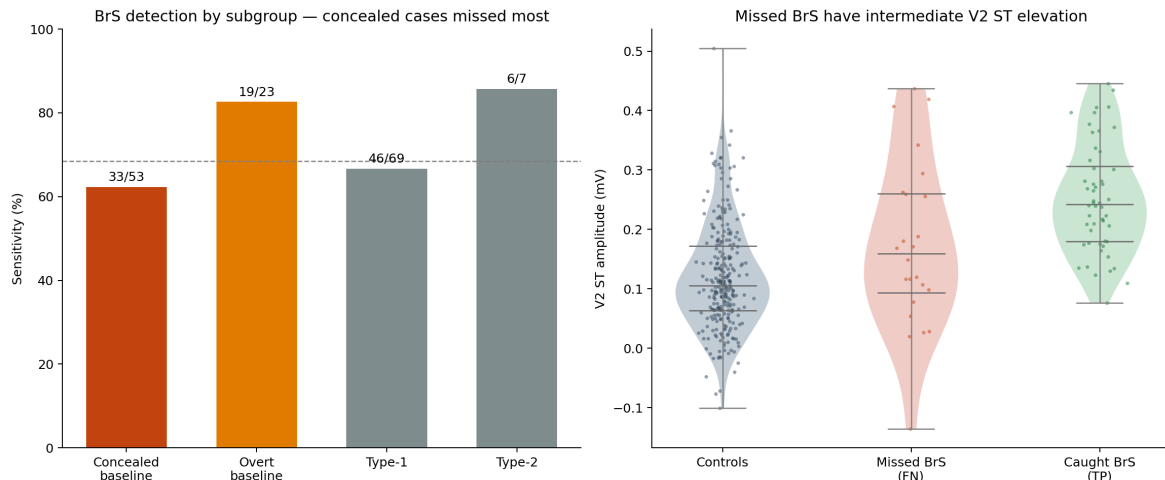


Figure 6: Left: sensitivity by subgroup. Concealed-baseline BrS cases are detected far less reliably than overt-baseline cases; the dashed line marks overall BrS sensitivity. Right: V2 ST amplitude by outcome group — missed BrS cases (false negatives) sit between controls and correctly caught BrS cases (true positives), rather than resembling either group cleanly.

5. Discussion

5.1 How this compares to prior BrS-AI work

Table 4 places this project’s numbers next to the other BrS-AI results discussed in Section 1.4. This is presented as context, not a leaderboard: the datasets, patient populations, provocation status, input signal type, and evaluation metrics all differ, so no row-to-row comparison in this table should be read as “better” or “worse” without that caveat attached.

Table 4. BrS-AI approaches in context (not directly comparable — see note below).

Study	Data / signal	Model	Headline result
Melo et al., 2023 [6]	Raw ECG waveform, no provocation	Deep neural network	88.4% accuracy, 0.934 AUC (validation)
Zanchi et al., 2023 [7]	P-wave morphology only, n=123	AdaBoost	81% accuracy, 86% sensitivity, 73% specificity
Ronan et al., 2025 [8]	Raw ECG, self-supervised pretraining	VICReg + classifier	AUROC 0.88, AUPRC 0.82
This work	Interpretable per-lead features, n=363	XGBoost	AUROC 0.90, 76% sens @ 90% spec

Note: AUROC/AUPRC and accuracy are not interchangeable, sample sizes range from 123 to 363, and provocation status of the underlying cohorts differs across studies. This table should be read as “the general shape of results in this sub-field looks similar” rather than a controlled benchmark.

What is fair to say is that an interpretable, feature-based model on a brand-new dataset lands in the same performance neighborhood as more complex deep-learning and self-supervised approaches built on other cohorts. That is a modest claim, and it is the honest one.

5.2 What the error analysis actually teaches

The most clinically meaningful result in this project is arguably not the AUROC — it is the fact that the model’s failure mode reproduces the real diagnostic problem instead of failing arbitrarily. A classifier that missed cases randomly would tell a clinician nothing useful about where to focus. A classifier whose failures concentrate on concealed-phenotype patients, with intermediate ST elevation between controls and clear cases, is behaving the way a cautious human reader might: uncertain exactly where uncertainty should exist. That is a testable, physiologically grounded failure mode rather than a black box shrugging.

5.3 Limitations

I want to be direct about what this project is not, both because it is good scientific practice and because a passion project loses credibility the moment it overstates itself.

1. **Single-center, single-dataset.** All data come from one hospital on one equipment configuration, with no external validation cohort. The reported metrics almost certainly overstate performance under any real domain shift — different ECG machines, different populations, different lead placement conventions. Cross-dataset testing against an independent BrS cohort is the single most important next step, not an optional one.
2. **Small sample.** 76 BrS patients, only 7 of them type-2, limits statistical power and widens every confidence interval reported here. The type-2 sensitivity figure (6/7) is not a real estimate; it is closer to an anecdote and is labeled as such throughout.
3. **Coarse sampling rate.** 100 Hz is low for fine QRS and J-point morphology; some ST-slope and notch descriptors are almost certainly quantized more than they would be at a clinical-grade sampling rate (500 Hz or higher). A higher-rate recording would likely sharpen exactly the features that matter most here.
4. **Diagnostic, not per-beat, labels.** The `basal_pattern` flag indicates that some confirmed BrS patients had a non-diagnostic resting ECG at the time it was recorded. That caps the sensitivity any resting-ECG-only classifier can achieve — it is a ceiling built into the problem, not a defect in this particular model.
5. **This is not a diagnostic device.** It is a research baseline, unvalidated externally, uncalibrated for deployment, and not reviewed by any regulatory body. It must not inform an actual clinical decision, and nothing in this paper should be read as suggesting otherwise.

6. Conclusion and next steps

This project set out to understand a specific clinical problem — why Brugada syndrome is so often missed on a resting ECG — and to see whether a transparent, feature-based machine-learning model could meaningfully engage with that problem rather than just fit a dataset. The resulting classifier reaches an AUROC of 0.90 and 76% sensitivity at 90% specificity on the first published baseline for the Brugada-HUCA dataset, with SHAP attribution confirming it relies on the clinically correct right-precordial signal, and an error analysis showing its failures concentrate — sensibly — on the same concealed-phenotype patients that challenge human readers and motivate provocation testing in the first place.

The most useful next steps, in rough priority order, are: (i) external validation on an independent BrS cohort to measure real domain shift, since everything reported here comes from a single hospital; (ii) a raw-waveform deep-learning comparator (e.g., a 1-D CNN or the self-supervised approach used by Ronan et al. [8]) benchmarked directly against this interpretable feature model on the same splits; (iii) a focused study of the concealed-phenotype subgroup, potentially incorporating the **sudden_death** outcome field that this project deliberately did not use; and (iv) threshold selection driven by the actual clinical cost of a missed case versus a false alarm, rather than the default 0.5 cutoff used for the headline confusion matrix here.

Data and code availability

Data: Brugada-HUCA, PhysioNet v1.0.0 [11]. Analysis was performed in Python using **wfdb**, **neurokit2**, **scikit-learn**, **XGBoost**, and **SHAP**. Feature tables, cross-validation metrics, effect-size statistics, and the per-patient error analysis referenced throughout this paper are included as supplementary CSV files alongside this manuscript.

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